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CLASS: BSCS- 6TH SEMESTER

SECTION: B

DEPARTMENT: COMPUTER SCIENCE

COURSE: ARTIFICIAL INTELLIGENCE

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ENGLISH TO URDU TRANSLATOR AGENT

Q: Identify the problem in the context of your agent.
Most of the time, when you are reading a book or watching a show in English, you come across words you don't understand. At that point, our agent can translate that word or sentence into Urdu accurately, so you can understand the meaning.

Q: Problem formulation. (Text, graphical representation)

Initial state: Raw English text/sentence that the user wants to translate (not yet translated).

Goal state: Accurate Urdu translation of the raw English text/sentence.

Processing:
1- Tokenization: Break the sentence into individual words.

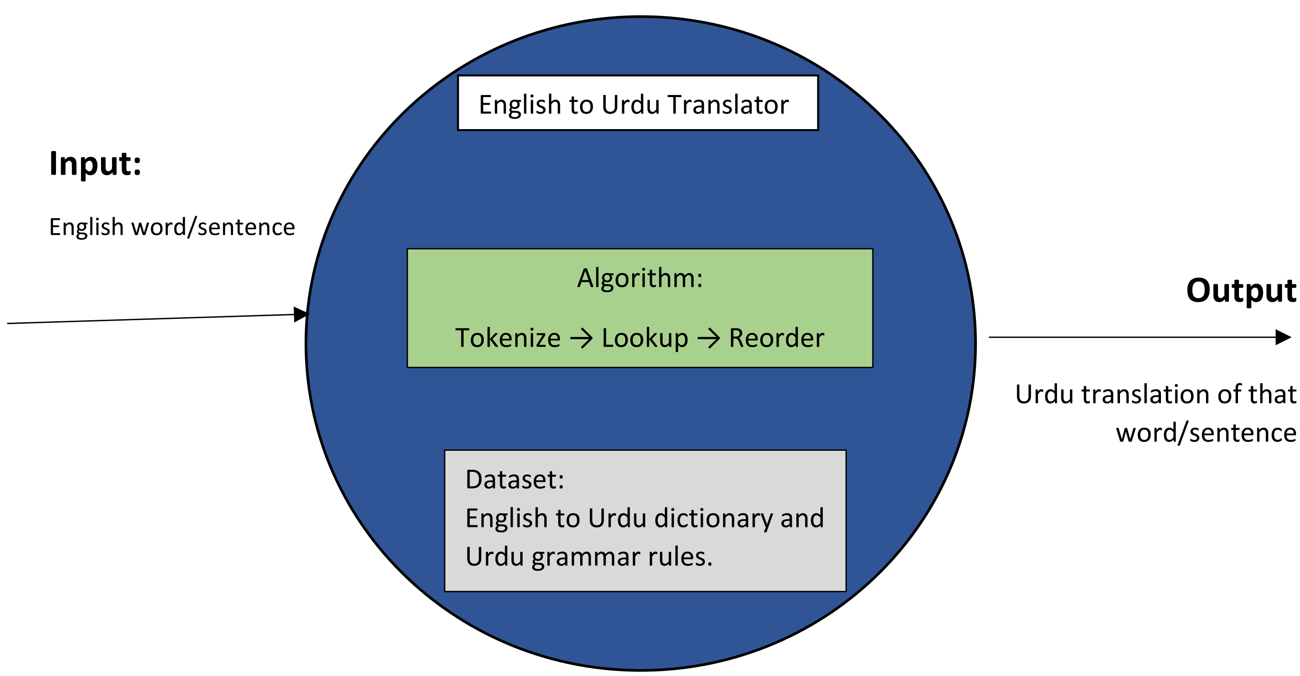
2- Translation lookup: Search the meaning of each word.

3- Reordering: Rearrange the translated words into correct Urdu sentence order.

Goal test:

Check if the Urdu sentence is grammatically correct and conveys the same meaning as the original English sentence.

Graphical Representation:



Q: Evaluate in PEAS.

Performance measure: Grammatical correctness and accuracy of the translated response.

Environment: User, input source (books, documents, shows), and the translation interface/application.

Actuator: Output screen showing Urdu translation.

Sensor: Text input field where user types English sentence/word.

Q: Identify the environmental properties of your agent.

Environmental properties:

Fully observable: because once the user types the sentence, the agent has the complete input text plus its full dictionary/grammar knowledge available immediately. Nothing is hidden from it.

Deterministic: because given the same English input, the agent will always produce the same Urdu translation (no randomness involved).

Episodic: because each sentence is translated independently; one translation doesn't depend on or affect the next one.

Static: because the input sentence stays the same while the agent is processing/translating it; nothing changes on its own during that time.

Discrete: because words and letters are countable, separate units (not a continuous range of values).

Known: because the agent has full knowledge of grammar rules, word meanings, and sentence structure rules needed to perform the translation.

Single-agent: because the translator works alone, using its own dictionary/rules to process input and produce output, without needing to interact with other agents.